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Mohamad Saad

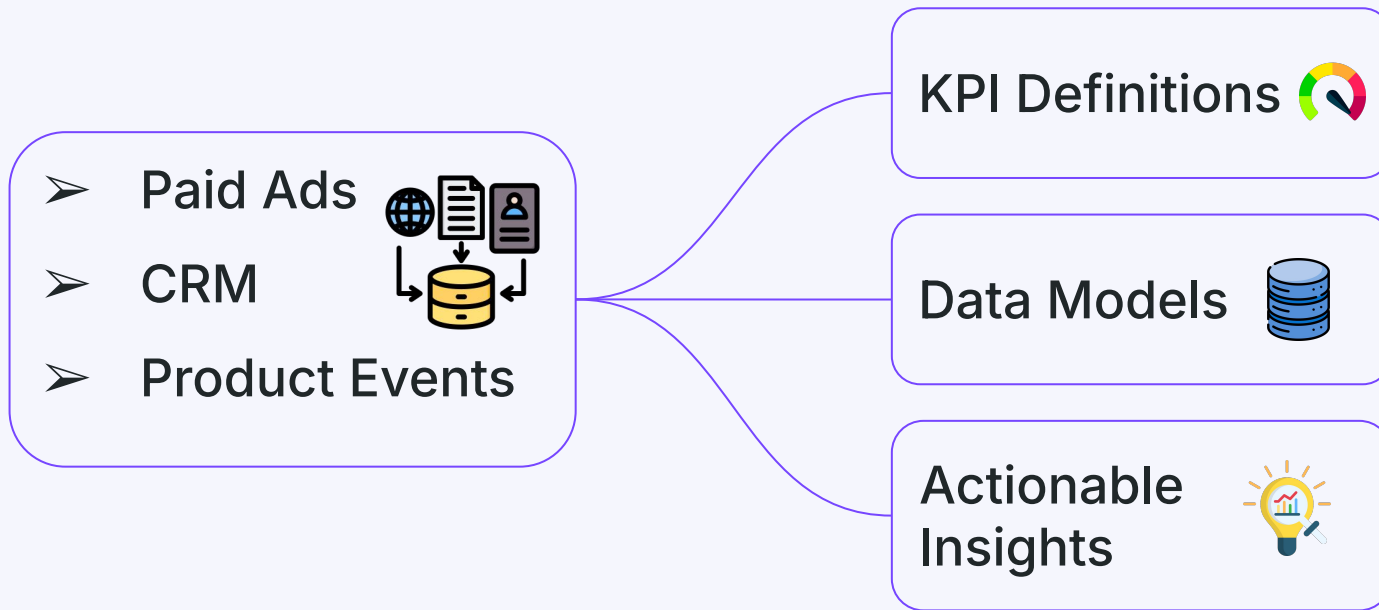
Case Study - Senior Growth Analyst



- 1) Introduction: introduce the problem and the case briefly
- 2) Data Overview: present the data as it is with some visuals
- 3) Data Quality & Assumptions: address data quality checks and stated assumptions
- 4) KPIs: choose and justify the 5 marketing metrics
- 5) Data Model: lay-out the proposed model for the data
- 6) Exploratory Insights: deeper dive into what the data shows
- 7) Recommendations: make the consequent recommendations

1- Introduction

We are charged with defining, organizing and analyzing the data provided as a fictional DeskBird case:



2- Data Overview

Three raw datasets capturing the customer journey from acquisition through conversion to retention:

Paid Ads

- 3 Channels
- 19 Campaigns
- ~\$1.56M in spend
- ~9 months

CRM

- 500 leads
- 120 companies
- ~\$450k in revenue*
- ~9 months

Product events

- ~33.5k events
- 115 companies
- ~5.6k unique users
- ~12 months

3) Data Quality & Assumptions

As per the ambiguous nature of the assignment, we note the following irregularities in the data that we found and assumptions we took:

1. Identity Resolution: only **120 unique real companies** (account-grain → company_name)
2. CRM data discarded: `<plan_tier>` & `<seats_used>` differ over the same account/user, seemingly at random. These **2 columns are not reliable** and will not be used for analyses.
3. CRM amounts: the `<amounts>` figure in the CRM table does not correlate with company size (`<seats_used>`) nor with `<tier_plan>` ($r = 0.07$ & 0.05 respectively).
4. Churn: A company has churned if, for a certain quarter, none of the accounts/users associated with it have induced any event.

4) KPIs

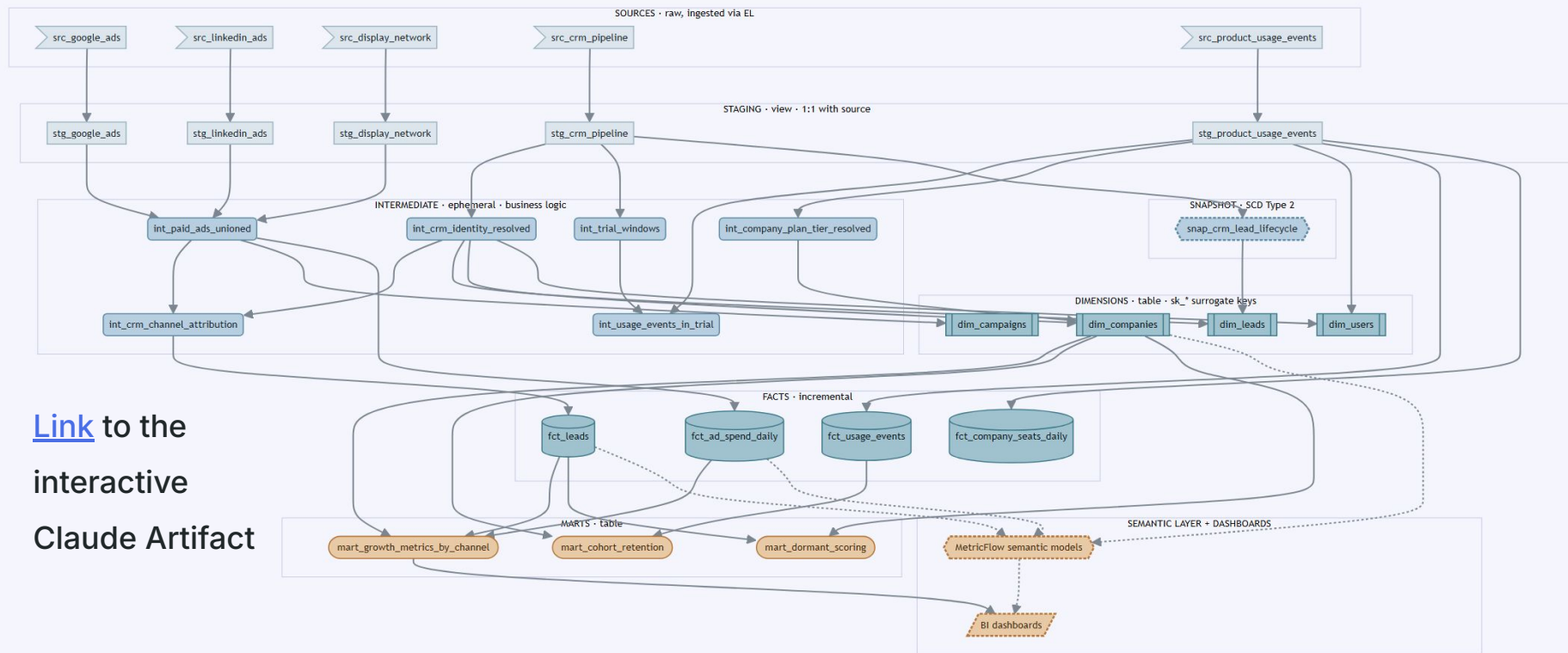
Our choice of metrics is driven by their utility to recommend actionable insights for different teams in the organization (Perf, Prod, Finance,...):

Metric	Rationale	Value	Note
CAC (Channel/blended)	Used to evaluate channel and overall marketing cost efficiency	\$31.8k blended (\$30.0k with organic)	Online marketing failing in general, left to see whether the weakness is structural or incidental
Conversion Rate (Unique Lead → closed won)	Post-lead process (sales) efficiency	43.3% win rate (55% of closed, 45% of inactive*)	Indicates a very strong post-lead process.
LTV:CAC (Channel/blended)	Ultimate efficiency metric (lifetime ROAS)	0.27 (paid only)	We are paying ~3.5x more to acquire a client than what they would bring us over their lifetime. For a healthy business, this would be between 3 & 5.
CPL (Channel)	North star for performance marketing	\$13.9k blended (Company grain)	Possible to give a monetary value to a lead (rather than a conversion) for better bidding.
Churn	How fast are our clients leaving?	%60.5 quarterly	IRL, this (and consequently our LTV calculations) would be widely unreliable.

*Inactive deals = Closed Deals + Dormant (See KM Analysis in Notebook)

5) Data Model

Full data model, from sources to marts as follows:



[Link](#) to the
interactive
Claude Artifact

5) Data Model (Cont'd)

Full data model, from sources to marts as follows:

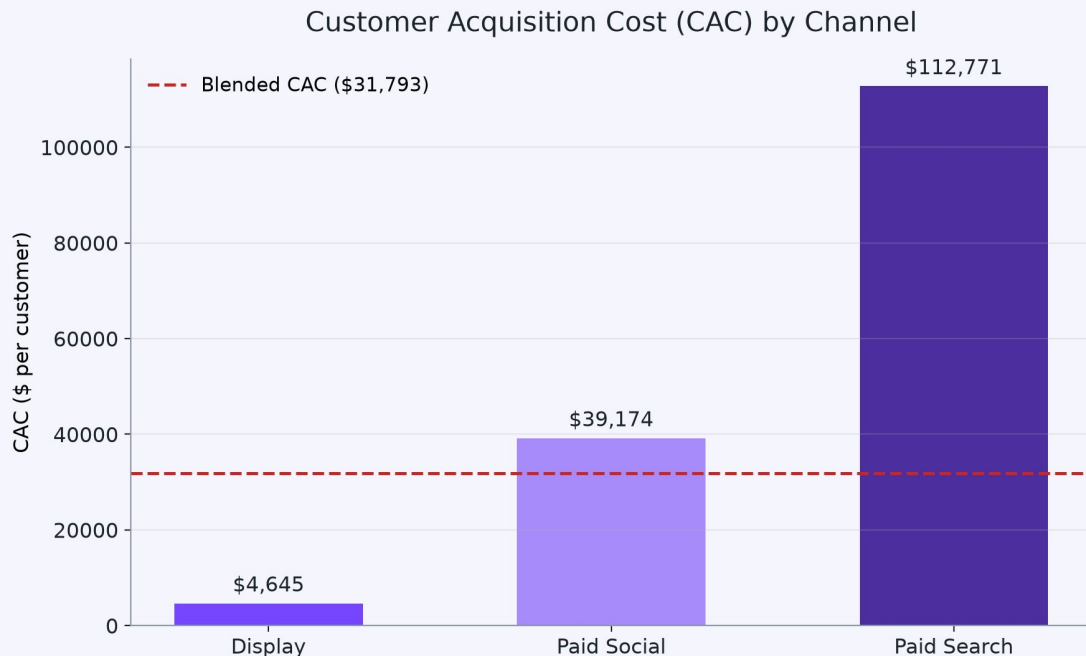
1. Layered dbt lineage: sources → staging → intermediate → facts + dims → marts
2. Staging: views, as they are cheap, always-fresh 1:1 mirrors of raw sources, with no business logic yet at this stage
3. Intermediate: ephemeral, this is where the data-quality findings from earlier in the analysis get encoded as reusable logic, not re-derived every time (*Caveat*: the CRM life-cycles table will use SCD2 to be able to track changes to statuses of leads & clients)
4. Dims: Use of tables with surrogate keys
5. Facts: merge, append and insert_overwrite depending on immutability and persistence of data (details tbd according to source specs)
6. Marts: materialized as tables, containing the 5 aforementioned KPIs and feeding the semantic layer used for dashboarding

The reader will check the accompanying Jupyter Notebooks for further details about materialization, testing and documentation

6) Exploratory Insights: Acquisition

Display shows lowest CAC by far (~\$4.6k per customer acquired). A fact, given the current setup, might be misleading:

- CAC is calculated by tracing the conversion to the origin lead's channel
- In case of multiple leads for the same company converting, attribution goes to the lead of the first conversion
- The apparent success of Display can be explained (and debunked) by looking at channels from an **incrementality perspective**



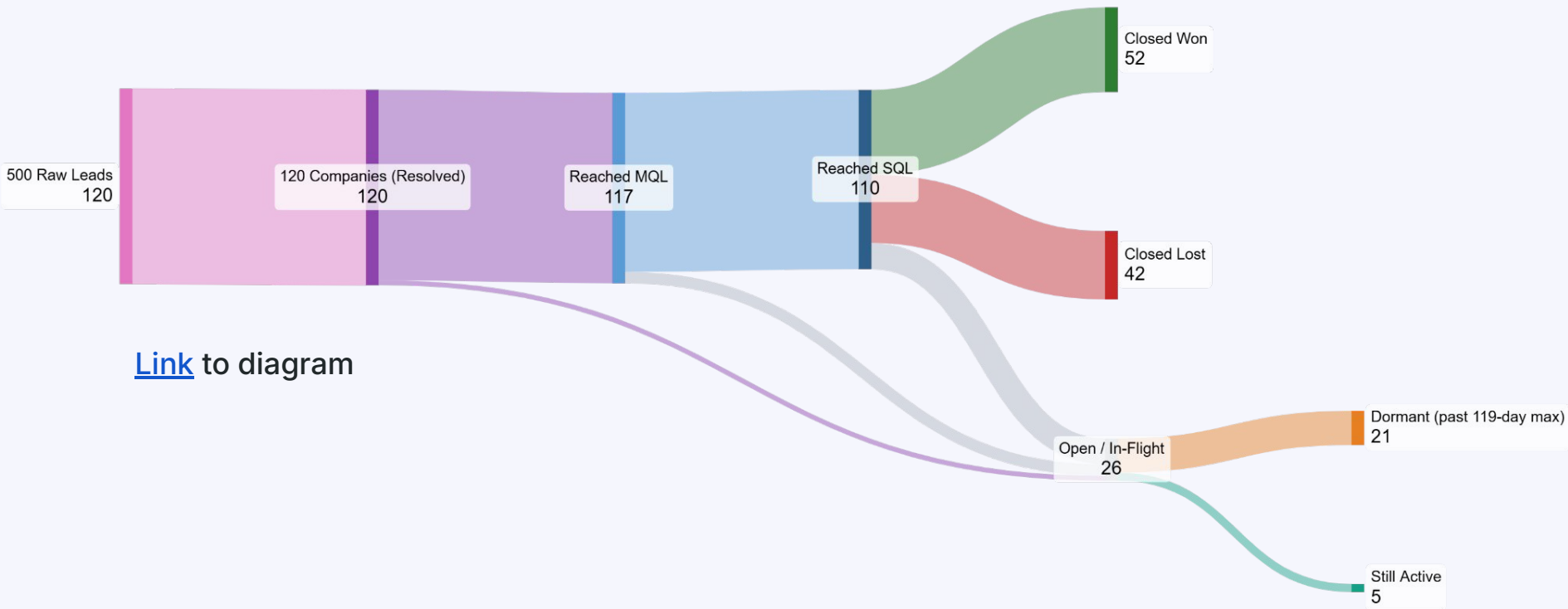
6) Exploratory Insights: Acquisition (Con't)

Display shows lowest CAC by far (~\$4.6k per customer acquired). A fact, given the current setup, might be misleading:

1. Display is generally used as a retargeting channel, re-engaging users who interacted with the brand elsewhere without converting, placing it structurally **closer to the end of the customer journey**.
2. Combined with last-touch attribution, it gets credited for conversions that other channels actually "caused" upstream. It is expected that the majority of **leads attributed to Display are not "incremental"** and therefore would have happened without it.
3. This is a structural **bias in the measurement** setup, not a data quality issue usually solved with a setup of multi-touch attribution model and a set of incrementality experiments to serve as (i) sources of truth and (ii) calibration parameters.
4. Before allocating budget toward Display: a geo-lift holdout test (Display off in some regions, on in others) would isolate its true incremental contribution rather than trusting attributed credit alone.

6) Exploratory Insights: Conversion

Of 500 leads, our funnel closed 52 companies, with 21 dormant and 5 still active:



6) Exploratory Insights: Conversion (Con't)

Of 500 leads, our funnel closed 52 companies, with 21 dormant and 5 still active:

1. The gap between the 500 leads and 120 companies is not treated as noise but as a funnel step that unifies the identity of the companies and avoids double counting
2. Kaplan-Meier survival analysis shows no company has ever taken longer than 119 days to close a deal. Applying that threshold to the currently-open pipeline, 21 of 26 open cases are deemed as "dormant". (Jupyter notebook for further details)
3. Metrics across the funnel seem to be widely over-performing industry benchmark (~43% win rate vs 20%-25% for [B2B SaaS industry benchmark](#))

6) Exploratory Insights: Retention

Quarterly cohort retention shows ~60.5% churn rate, but rests on a thin, unbalanced set of cohorts:

- We use quarterly cohorts as opposed to monthly as the latter proved extremely thin & unusable sets
- ~60.5% churn is the average period-1 retention across the three fully-observed cohorts (33%, 32%, 53%), treated as a constant quarterly rate

Quarterly Cohort Retention — share of each cohort active N quarters after close
(orange rows = right-censored, excluded from headline churn figure)

	Period -1*	Period 0	Period 1	Period 2	Period 3	Period 4
2025Q1 (n=6)		100%	33%	33%	33%	17%
2025Q2 (n=22)	77%	77%	32%	18%		
2025Q3 (n=15)	60%	93%	53%			
2025Q4 (n=8)	88%	88%	25%			
2026Q1 (n=1)	100%					

*Period -1 = pre-close usage (e.g. trial activity), shown for context; retention itself starts at Period 0 (the close quarter).

6) Exploratory Insights: Retention (Con't)

Quarterly cohort retention shows ~60.5% churn rate, but rests on a thin, unbalanced set of cohorts:

1. Churn/retention is measured by tracking each Closed-Won cohort's continued product usage in the quarters following close. A company counts as "retained" in a period if it shows any usage event that quarter.
2. 2025Q4 & 2026Q1 cohorts are excluded from that average (right-censored), since usage data only runs to mid-January 2026, not enough time has passed to observe their true retention.
3. Cohort sizes are small and uneven (6 to 22 companies per quarter). The reader will take this churn figure as directional, not a statistically stable estimate. In fact the first period of the churn curve is usually the steepest before it stabilizes at "steady-state".

7) Recommendations

Based on all that proceeded, our observations and the extensive details in the main analysis presented in the Jupyter Notebook, we make the following recommendations:

- Fixing identity resolution issue, preferably at the source where the CRM can immediately output company-level data
- Switch to multi-touch attribution with an emphasis on incrementality testing for truth and calibration
- Disqualify or reactivate lead after the 119-day mark as considered to be statistically dormant
- Standardize the *<amount>* entry in the CRM where prices do not correlate with seats or plan tiers at all
- Investing in retention efforts, specifically during the first quarter post-win-close
- Reconsider the 14-day trial: a discounted first quarter may convert and retain better for a team-adoption product (to which teams might take a while to get used)

Thank You!

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